

WattWise · Industrial Energy Intelligence Platform ·

Cut industrial energy costs by changing a single machine

An AI-powered energy arbitrage and load optimization platform that turns Pakistan's load shedding crisis into a competitive advantage for industrial users.

Rs. 5M

avg. monthly savings per mill

4–8 wk

hardware installation time

5,000

target industrial machines per mill

01 Overview & Vision

WattWise is a B2B SaaS + IoT platform that installs non-intrusive smart current sensors on industrial machinery and uses machine learning to optimize when heavy processes run — automatically shifting energy load to cheap grid hours, pre-staging generator switchovers, and surfacing real-time cost intelligence to factory managers.

Pakistan's industrial sector burns an estimated **Rs. 180–250 billion annually** in excess energy costs from unmanaged load shedding. Factories run diesel generators at Rs. 80–100/kWh when grid power costs Rs. 22–40/kWh — a 2–4x price multiplier that nobody is managing intelligently. WattWise closes that gap.



Core thesis: We don't sell energy. We sell the intelligence to use it better. Our sensor hardware is the wedge; our recurring SaaS revenue is the business. We take 20% of documented savings — zero upfront cost to the factory.

Mission Statement

To make Pakistani industry globally competitive by eliminating the energy cost disadvantage through AI-driven load intelligence — one factory at a time, starting with Punjab's textile belt.

Company Structure (Proposed)

LEGAL ENTITY

Private Limited Company registered under SECP, Islamabad. Technology company classification for tax incentives under SRO 489(I)/2007 (IT Export Policy). Target PSEB registration within 6 months for export exemptions.

INCORPORATION TIMELINE

SECP registration takes 3–5 business days online. NTN registration: 2–3 days via FBR IRIS. Bank account (Meezan Bank or HBL recommended for startup-friendly tier): 1 week. Total: **~2 weeks to operational entity.**

02 Problem Statement

The Energy Crisis in Numbers

18–22 hours/day average load shedding in industrial feeders (2024– 2026)	3.5× cost of diesel generation vs. grid power per kWh	Rs. 2.8T Pakistan's annual energy subsidy bill — unsustainable and shrinking	0 industrial units currently using AI for load management
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Root Causes

 Zero Visibility	 Manual Generator Switching
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Factory managers have no real-time view of energy consumption by machine, line, or shift. WAPDA bills arrive monthly and tell you nothing actionable.

Switchover decisions are made by a chowkidar or electrician on duty. Lag time of 3–15 minutes per event = machines idle, orders delayed, quality drops.



Unpredictable Schedules

WAPDA load shedding schedules are publicly posted but not systematically tracked. No industrial unit is using them to pre-schedule energy-intensive processes.



Flat-Rate Generator Runtime

Generators run whether machines are at peak load or idle. No factory measures generator efficiency per productive output unit. Waste is invisible.



Quality Hits During Transitions

Voltage fluctuations during grid↔generator switchover cause dyeing batch failures, loom thread breaks, and CNC calibration drift — unmeasured losses.



No Data for ESCOs or Lenders

Banks and Energy Service Companies (ESCOs) can't finance energy upgrades because there's no verified baseline consumption data to underwrite against.



The opportunity gap: A medium-sized textile mill in Faisalabad consumes 400–800 kWh/hour. Running on generator at Rs. 90/kWh vs. grid at Rs. 30/kWh = Rs. 24,000–48,000/hour waste. That's **Rs. 288,000–576,000 per 12-hour shift** in avoidable cost — on just one factory.

Who Suffers Most

Sector	Pakistan Employment	Annual Energy Waste Est.	Pain Level
Textile & Apparel	15M+ workers	Rs. 80–120B	CRITICAL
Surgical & Medical Instruments	50,000+ (Sialkot)	Rs. 8–12B	HIGH

Leather & Tanneries	500,000+	Rs. 15–20B	HIGH
Rice & Food Processing	2M+	Rs. 12–18B	MEDIUM
Pharmaceuticals	100,000+	Rs. 6–9B	MEDIUM
Steel Re-rolling	300,000+	Rs. 20–30B	CRITICAL

03 Product Solution

WattWise is a three-layer product: a **hardware sensor network** that reads real-time load without rewiring, a **cloud intelligence platform** that models consumption and predicts optimal schedules, and a **factory floor dashboard** that turns predictions into automated decisions.

Product Components



WattClamp™ Sensor Node

Clip-on current transformer (CT) that attaches to existing cables in 5 minutes. No electrician license needed. Sends real-time ampere readings via WiFi/4G/LoRa.



WattBrain™ Edge Controller

On-site edge compute unit (Raspberry Pi CM4 class) that runs local ML inference, controls relay switches for non-critical loads, and maintains 72h offline buffer.



WattCloud™ SaaS Platform

Web + mobile dashboard for factory managers. Shows real-time consumption, cost per kWh, savings tracker, schedule recommendations, and compliance reports.



LoadShift™ Scheduler

AI engine that ingests WAPDA feeder schedules, historical load patterns, and production plans to output an optimized process schedule — daily, auto-updated.



SwiftSwitch™ Automation

Automated relay control module that triggers generator switchover 8–12 seconds before predicted load shedding — before the grid drops, not after.



SavingsLedger™ Reports

Auditable, tamper-proof savings documentation per billing period. Used for our revenue-share invoicing and can be shared with banks for ESCO financing.

The Value Proposition — Explained Simply



Before WattWise: Load goes out → chowkidar sees lights off → runs to generator room → starts generator → 8–15 minutes of machines idle or running on bad power → textile defects, rejected batches, production loss.

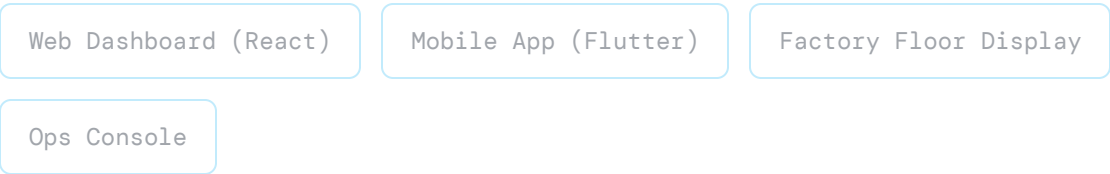
After WattWise: System predicts load shedding 10 minutes in advance → heavy dyeing vats automatically start pre-heating cycle → non-critical AC units throttled → generator starts cold → seamless switchover in 8 seconds → zero production loss.

04 System Architecture

WattWise uses a three-tier architecture: an **edge sensing layer** deployed on-site, a **cloud processing layer** hosted on AWS (Bahrain region for data sovereignty), and a **presentation layer** accessible via web and mobile apps.

WATTWISE SYSTEM ARCHITECTURE

LAYER 4 – PRESENTATION



↕ REST / WebSocket / gRPC

LAYER 3 – CLOUD PLATFORM (AWS BAHRAIN)



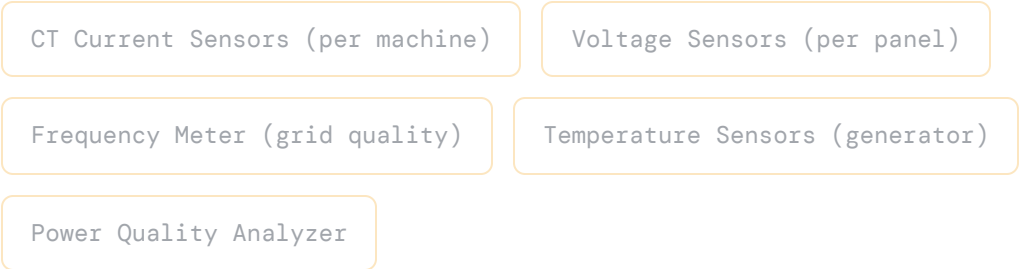
↕ MQTT over TLS / HTTPS

LAYER 2 – EDGE (ON-SITE: WATTBRAIN™ CONTROLLER)



↕ Modbus / Custom Serial / Digital GPIO

LAYER 1 – PHYSICAL SENSING (WATTCLAMP™ NODES)



Key Design Principles

- ✓ **Offline-first edge:** WattBrain stores 72 hours of data locally and runs critical automation independently. A cloud outage never stops factory protection.
- ✓ **Non-invasive sensing:** All sensors clip onto existing cables via split-core CTs. No electrician needed, no production downtime for installation.

- ✓

Multi-network resilience:

WiFi primary, 4G SIM failover, LoRa backup for remote factory areas. Data always reaches the cloud.
- ✓

Data sovereignty:

All customer data stored on AWS ME-South-1 (Bahrain). Pakistani customers don't need to worry about cross-border data concerns.
- ✓

Zero-trust security:

Per-device TLS certificates, JWT auth, rate limiting on all APIs, encrypted at-rest with AES-256.

05

Hardware Layer

WattClamp™ Sensor Node Specifications

Parameter	Specification	Notes
Sensor Type	Split-core Current Transformer (CT)	No cable cutting required
Current Range	0–600A (models: 0–50A / 0–200A / 0–600A)	Covers most industrial machinery
Accuracy	±0.5% of full scale	Class 0.5 IEC 61869–2
Voltage Measurement	0–690V AC (3-phase)	Panel-mounted voltage tap
Sampling Rate	10 kHz per channel	Captures harmonics and transients
Connectivity	RS-485 Modbus to WattBrain	Up to 32 nodes per bus segment
Power Supply	Self-powered via CT + 5V DC backup	No separate wiring needed
Enclosure Rating	IP54	Suitable for dusty textile environment
Operating Temp	–10°C to +70°C	Handles Pakistan summer factory

floors

Unit Cost (BOM)	~USD 12–18 per node	Sourced from Shenzhen via LC or agent
Retail/Customer Price	Free (bundled in subscription)	Customer never pays for hardware upfront

WattBrain™ Edge Controller Specifications

Parameter	Specification
Compute Module	Raspberry Pi CM4 (4GB RAM, 32GB eMMC) or Rockchip RK3566 equivalent
OS	Ubuntu 22.04 Server (ARM64), WattWise Edge OS layer
ML Runtime	ONNX Runtime 1.16+ with optional TensorRT acceleration
Local Storage	32GB eMMC + 128GB SD backup
Communication	Ethernet (primary), WiFi 5 (backup), 4G LTE via USB dongle
Relay Outputs	8× digital relay outputs (24VDC/10A) for load control
Serial Ports	4× RS-485 for CT node buses (128 nodes total)
Display Port	HDMI for optional factory floor dashboard screen
Offline Autonomy	72 hours local data buffer + full automation capability
BOM Cost	~USD 120–160 per unit
Deployment	1 per factory section / MDB panel (typical factory: 2–4 units)

SwiftSwitch™ Pre-emptive Relay Module

This is the hardware that delivers the "8-second switchover" — a key differentiator. The module connects between the factory's ATS (Automatic Transfer Switch) control signal and the WattBrain, intercepting and pre-empting the switchover command based on our prediction engine rather

than waiting for grid voltage collapse detection.



Safety Note: The SwiftSwitch module is installed by a WattWise-certified electrician only. It is rated for 660V AC, 63A, with fail-safe mechanical relays that default to GRID in case of microcontroller fault. Certifications required: PSQCA, IEC 60947-1.

Procurement Strategy

Phase 1 (0–500 units): Source fully assembled from Shenzhen via Alibaba Trade Assurance + LC. Local customs duty: CT sensors fall under HS Code 8504.10 (transformers), currently at 5% + 3% sales tax. Phase 2 (500+ units): Explore local assembly with LUMS NIC / PITB hardware incubation support. Phase 3 (2,000+ units): OEM manufacturing in Pakistan (Sialkot or Lahore industrial zones) targeting PSQCA certification for local quality mark.

06 Software Stack

Technology Choices & Rationale

Layer	Technology	Why This Choice
Edge Runtime	Python 3.11 + ONNX Runtime	ONNX allows model export from any framework; runs on ARM64 without GPU
Edge Broker	Eclipse Mosquitto (MQTT 5.0)	Sub-100ms latency; works on 2G connectivity; industry standard for IoT
API Backend	Go 1.22 (main), Python FastAPI (ML)	Go for high-throughput sensor ingestion; Python for ML serving
Message Queue	Apache Kafka (MSK on AWS)	Handles 10M+ sensor readings/day; replay capability for retraining
Time-Series DB	InfluxDB v3 (Cloud)	Purpose-built for high-frequency numeric sensor data with downsampling
Relational DB	PostgreSQL 16 (RDS)	Tenant config, billing, user management, savings

	Aurora)	leager
Cache Layer	Redis 7 (ElastiCache)	Real-time dashboard reads; rate limiting; session management
ML Training	Python, PyTorch, scikit-learn, Prophet	Prophet for time-series forecasting; PyTorch for custom load models
ML Deployment	SageMaker + ONNX export to Edge	Train in cloud, deploy lightweight model to WattBrain
Frontend Web	React 19 + TypeScript + Recharts	Recharts for energy visualization; fast iteration on dashboards
Mobile App	Flutter 3.x (iOS + Android)	Single codebase; Urdu font support (Google Noto Nastaliq)
Infrastructure	AWS (Bahrain me-south-1)	Closest region; GDPR-adjacent; SLA 99.99%
IaC	Terraform + GitHub Actions	Reproducible infra; CI/CD on every PR
Monitoring	Grafana + Prometheus + CloudWatch	Full observability stack; alert on sensor drop-out

Edge Software Architecture (WattBrain)

WATTBRAIN/MAIN.PY – EDGE PROCESS MANAGER

```
# WattBrain Edge OS – Core Process Manager
# Runs as systemd service on Ubuntu 22.04 (ARM64)

import asyncio
from wattbrain.sensor_bus import ModbusCTBus
from wattbrain.inference import LoadPredictor
from wattbrain.mqtt_client import WattMQTT
from wattbrain.relay_ctrl import RelayController
from wattbrain.local_store import RingBuffer

SAMPLE_INTERVAL_MS = 100    # 10 Hz per channel
PUBLISH_INTERVAL_S = 5      # Aggregate + send every 5s
PREDICTION_WINDOW = 900    # 15-min rolling inference window
```

```

async def main_loop():
    bus = ModbusCTBus(ports=["/dev/ttyUSB0", "/dev/ttyUSB1"])
    predictor = LoadPredictor.from_onnx("models/loadshift_v2.onnx")
    mqtt = WattMQTT(broker="mqtt.wattwise.pk", tls=True)
    relay = RelayController(gpio_map=RELAY_CONFIG)
    buffer = RingBuffer(max_hours=72)

    while True:
        readings = await bus.read_all()    # <5ms
        buffer.append(readings)

        if buffer.seconds_elapsed() % PUBLISH_INTERVAL_S == 0:
            agg = buffer.aggregate_last(seconds=PUBLISH_INTERVAL_S)
            await mqtt.publish(f"factory/{FACTORY_ID}/telemetry", agg)

        prediction = predictor.infer(buffer.last(PREDICTION_WINDOW))
        if prediction.grid_drop_confidence > 0.85:
            relay.execute(prediction.recommended_actions)

        await asyncio.sleep(SAMPLE_INTERVAL_MS / 1000)

```

Cloud Sensor Ingestion Pipeline

INGESTION/HANDLER.GO — HIGH-THROUGHPUT MQTT CONSUMER

```

// WattWise Cloud — Sensor Ingestion Service (Go)
// Handles ~50,000 sensor readings/second at scale

package ingestion

type TelemetryPacket struct {
    FactoryID string    `json:"factory_id"`
    NodeID    string    `json:"node_id"`
    Timestamp time.Time  `json:"ts"`
    CurrentAmps float64  `json:"i_rms"`
    VoltageV   float64  `json:"v_rms"`
    PowerW     float64  `json:"power_w"`
    PowerFactor float64  `json:"pf"`
    GridSource bool      `json:"on_grid"`
}

func (h *Handler) IngestBatch(packets []TelemetryPacket) error {
    points := make([]influxdb.Point, len(packets))
    for i, p := range packets {
        points[i] = influxdb.NewPoint(

```

```
points[i] = influxdb.NewPoint(  
    "sensor_reading",  
    map[string]string{"factory": p.FactoryID, "node": p.NodeID},  
    map[string]interface{}{  
        "i_rms": p.CurrentAmps,  
        "power_w": p.PowerW,  
        "on_grid": p.GridSource,  
    },  
    p.Timestamp,  
)  
}  
return h.influx.WritePoints(context.Background(), points)  
}
```

07 API Reference

WattWise exposes a REST API (v1) for all dashboard and integration use cases. All endpoints require JWT Bearer authentication. Base URL: <https://api.wattwise.pk/v1>

Authentication

POST`/auth/token`

Get access token

```
{  
  "client_id": "ww_c_XXXXX",  
  "client_secret": "ww_sk_XXXXX",  
  "grant_type": "client_credentials"  
}  
// Response: { "access_token": "eyJ...", "expires_in": 3600 }
```

Real-Time Telemetry

GET`/factories/{factory_id}/telemetry/live`

Current sensor readings (WebSocket available)

```
// Response  
{
```

```
"factory_id": "fsd_mill_001",
"timestamp": "2026-10-05T09:15:32Z",
"grid_status": "ON",
"total_kw": 847.3,
"cost_per_hour_pkr": 25414,
"active_source": "GRID",
"nodes": [
  { "id": "node_01", "label": "Loom Section A", "kw": 312.1 },
  { "id": "node_02", "label": "Dyeing Vats", "kw": 284.7 }
]
```

Load Prediction

GET /factories/{factory_id}/predictions/schedule

Today's optimized load schedule

```
// Response
{
  "date": "2026-10-05",
  "predicted_outages": [
    { "start": "11:00", "end": "13:30", "confidence": 0.91 },
    { "start": "18:00", "end": "20:00", "confidence": 0.78 }
  ],
  "recommendations": [
    { "action": "RUN_HEAVY_LOAD", "from": "07:00", "to": "10:45", "reason": "Grid stable" },
    { "action": "PRE_HEAT_VATS", "at": "10:48", "reason": "Outage predicted at 11:00" }
  ],
  "estimated_saving_pkr": 184000
}
```

Savings Ledger

GET /factories/{factory_id}/savings?month=2026-09

Verified savings for billing period

```
{
  "period": "2026-09",
  "baseline_cost_pkr": 18200000,
  "actual_cost_pkr": 12940000,
  "gross_saving_pkr": 5260000,
  "wattwise_fee_pkr": 1052000,
  "net_saving_pkr": 4208000
}
```

```
    "roi_multiple": 4.0,  
    "audit_hash": "sha256:a3f8..."  
}
```

08 ML / AI Engine

WattWise's intelligence stack is composed of four specialized ML models that work together: a **load forecasting model**, a **grid outage prediction model**, a **process schedule optimizer**, and an **anomaly detector**.

Model 1: Grid Outage Predictor (GOP)

OBJECTIVE

Predict grid outages on a specific WAPDA feeder with a 10–15 minute horizon and $\geq 85\%$ confidence threshold before triggering automation.

INPUT FEATURES

- Historical load shedding schedule for feeder (scraped from WAPDA regional sites)
- Day-of-week, hour, season, Ramadan flag, public holiday flag
- Real-time grid voltage & frequency (leading indicators of impending trip)
- National generation deficit (scraped from NEPRA daily reports)

ARCHITECTURE

Gradient Boosted Trees (XGBoost) for schedule-based features + LSTM for real-time voltage/frequency signal. Ensemble output via calibrated probability.

TARGET PERFORMANCE

Metric	Target
--------	--------

Precision	>88%
Recall	>80%
False positive rate	<5% (no unnecessary generator starts)
Latency (edge)	<200ms inference

Model 2: LoadShift Optimizer (LSO)

Given today's predicted outage windows and the factory's production plan, LSO solves a **constrained optimization problem**: maximize production output while minimizing generator-hour consumption, subject to process constraints (dyeing cycles can't be interrupted, loom sections have minimum runtime requirements, cooling must maintain temperature bands).

Implemented as a Mixed Integer Linear Program (MILP) using Python's `scipy.optimize` or PuLP for smaller problems, and Google OR-Tools for multi-section factories. Solves in <3 seconds for a typical 20-process factory.

Model 3: Consumption Baseline Estimator

Uses Facebook Prophet (time-series decomposition) to establish each factory's energy consumption baseline — what they *would have spent* without WattWise. This baseline is the denominator in our savings calculation and is the key to our revenue model being auditable and dispute-free. Baseline is locked per month on the 1st and cannot be retroactively adjusted.

Model 4: Anomaly Detector

An unsupervised Isolation Forest model running on-edge that detects: unexpected consumption spikes (possible equipment fault or energy theft), sensor dropout (tampered or failed CT), and generator efficiency degradation (rising kWh/unit ratio over time). Alerts sent via SMS (Telenor/Jazz) and push notification within 90 seconds of detection.



Training Data Strategy: Cold start problem is solved by a *shadow mode* deployment — we install sensors 30 days before activating automation, purely collecting data. This builds a factory-specific model before any savings claims are made. It also builds trust with the factory owner.

09

Core Features

Dashboard — Factory Manager View

- ✓

Live Power Map:

Visual factory floor map with real-time kW consumption per section. Color-coded by load level (green/amber/red).
- ✓

Today's Outage Calendar:

Timeline showing predicted load shedding windows with confidence bands, overlaid on the production schedule.
- ✓

Generator Runtime Counter:

Real-time diesel consumption tracker with Rs. cost per hour vs. if grid were available.
- ✓

Savings Ticker:

Running total of savings this month vs. same period last year. The number factory owners love.
- ✓

Shift Reports:

Automated per-shift energy report emailed to factory manager. In Urdu upon request.

Automation — SwiftSwitch Behaviors

- ⚡

Predictive Pre-switch:

Start generator cold 8–12 minutes before outage, not after. Eliminates the warm-up lag.
- ⚡

Load Shedding:

Auto-switch off non-critical loads (HVAC, lighting in storage areas, water pumps) during generator operation.
- ⚡

Grid Return Detection:

Detect stable grid return (10-second voltage stability check) before switching back — prevents hunting on unstable grid.
- ⚡

Process Protection:

Never interrupt a protected process (dyeing vat, furnace, autoclave) mid-cycle, even if it means running on generator longer.

Reporting & Compliance

- 📄

Monthly Savings Certificate:

Auditable PDF with SHA-256 hash of underlying data. Can be shared with banks for ESCO financing applications.

-  **Carbon Emission Report:** kgCO₂ avoided by reducing diesel generation. Useful for EU buyer sustainability questionnaires (GSP+).
-  **WAPDA Bill Reconciliation:** Upload your WAPDA bill; we annotate each line item with our consumption data to catch billing errors (common in Pakistan).
-  **Generator Health Trend:** Tracks efficiency degradation over time, predicts maintenance windows before costly breakdowns.

10 Data Models

POSTGRESQL SCHEMA — CORE ENTITIES

```
-- Multi-tenant factory registry
CREATE TABLE factories (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  name        TEXT NOT NULL,
  sector      TEXT CHECK(sector IN ('TEXTILE', 'SURGICAL', 'FOOD', 'PHARMA', 'STEEL')),
  city        TEXT,
  wapda_feeder TEXT,          -- feeder ID for outage scraping
  plan        TEXT DEFAULT 'STARTER',
  created_at  TIMESTAMPTZ DEFAULT NOW()
);

-- Sensor node registry per factory
CREATE TABLE sensor_nodes (
  id          UUID PRIMARY KEY,
  factory_id  UUID REFERENCES factories(id),
  label       TEXT,          -- e.g. "Loom Section A"
  ct_range_a  INTEGER,       -- 50 / 200 / 600
  phase       INTEGER CHECK(phase IN (1,3)),
  installed_at TIMESTAMPTZ,
  last_seen   TIMESTAMPTZ
);

-- Savings ledger (immutable, append-only)
CREATE TABLE savings_records (
  id          UUID PRIMARY KEY,
  factory_id  UUID REFERENCES factories(id),
  period_month DATE,
```

```

baseline_pkr    NUMERIC(14,2),
actual_pkr      NUMERIC(14,2),
gross_saving_pkr NUMERIC(14,2),
fee_pkr         NUMERIC(14,2),
audit_hash      CHAR(64),           -- SHA-256 of raw data snapshot
locked_at       TIMESTAMPTZ,
UNIQUE(factory_id, period_month) -- one record per factory per month
);

```



Audit integrity: The savings_records table is append-only at the application level. Once locked, records can only be read — never updated. The audit_hash is a SHA-256 of the raw sensor data archive for that month, stored separately in S3 Glacier. This makes savings figures independently verifiable by third parties (banks, auditors).

11 Go-to-Market Strategy

Beachhead Market: Faisalabad Textile Cluster

Faisalabad is Pakistan's textile capital — 5,000+ textile units, 150,000+ looms, and some of the country's worst industrial load shedding. It is densely clustered (most mills within 25km radius), meaning field support cost is low and word-of-mouth is extremely powerful. One satisfied mill owner talks to 20 others at the APTMA (All Pakistan Textile Mills Association) monthly meeting.



Beachhead ICP (Ideal Customer Profile): Textile mill, 200–2,000 looms, monthly WAPDA bill Rs. 2M–15M, runs a generator 8+ hours/day, exports to EU/US buyers (sustainability reporting pressure). Medium-to-large family-owned business where the factory owner is also the decision-maker. Age 35–60. Cares about margin, not technology for its own sake.

Sales Motion

- 1 **APTMA** Approach APTMA Faisalabad chapter for a pilot program endorsement. Industry **Partnership:** association credibility eliminates cold-call resistance entirely. Offer 3 free pilot installations to association members.

- 2

Zero Upfront Offer:

"No hardware cost. No subscription. We take 20% of what we save you. If we save you nothing, you pay nothing." This removes every sales objection for the first 12 months.
- 3

30-Day Shadow Trial:

Install sensors in monitoring-only mode. After 30 days, show the factory owner a report of what they spent vs. what they could have spent. The data sells itself.
- 4

Customer Success Reps:

Hire 2 field CSRs based in Faisalabad from day one. Must speak Punjabi, understand textile operations. These people build the referral network.
- 5

Case Study Machine:

After first 5 customers, produce a one-page Urdu-language case study with the actual savings numbers (with customer permission). Distribute at APTMA, LCCI, and WhatsApp industry groups.

Expansion Sequence (Geographic)

Phase	City / Cluster	Sector Focus	Timeline
Phase 1	Faisalabad	Textile (weaving, dyeing, finishing)	Months 1–12
Phase 2	Sialkot	Surgical instruments, sports goods, leather	Months 9–18
Phase 3	Lahore	Mixed industry, pharmaceuticals, food processing	Months 15–24
Phase 4	Karachi	Steel, chemicals, large-scale manufacturing	Months 20–30
Phase 5	Gujranwala, Gujrat	Fan manufacturing, ceramics, steel furniture	Months 24–36

12 Pricing Model

WattWise operates on a **Gain-Share model** for customer acquisition (no upfront cost) and

STARTER (GAIN-SHARE)	★ GROWTH (ANNUAL SAAS)	ENTERPRISE
20% of monthly verified savings — no cap ✓ Up to 20 sensor nodes ✓ WattBrain edge unit (1 unit) ✓ LoadShift predictions ✓ Basic dashboard ✓ Monthly savings report ✓ WhatsApp alerts	Rs. 85K per month, billed annually ✓ Up to 64 sensor nodes ✓ 2x WattBrain units ✓ SwiftSwitch automation ✓ Full LoadShift optimizer ✓ Multi-section floor map ✓ Savings Certificate PDF ✓ Carbon report (EU export) ✓ Dedicated CSR	Custom negotiated per large account ✓ Unlimited sensor nodes ✓ Multi-site deployment ✓ Custom ML model tuning ✓ API access for ERP integration ✓ On-site quarterly reviews ✓ SLA: 99.9% uptime ✓ Bank-grade savings audit

Unit Economics (Growth Plan — Typical Customer)

Item	Amount (PKR)	Notes
Monthly Subscription Revenue	Rs. 85,000	Per customer, annual contract
Hardware BOM Cost (amortized 3yr)	Rs. (12,000)	~USD 120 WattBrain + 20 nodes × USD 15
Cloud Infrastructure	Rs. (4,500)	AWS per-customer allocation
Field Support (CSR allocation)	Rs. (8,000)	1 CSR per 25 customers
Gross Margin per Customer/Month	Rs. 60,500	~71% gross margin at scale

13

Financial Projections

Revenue Model — 36-Month Forecast

Month	Active Customers	MRR (PKR)	ARR (PKR)	Cumulative Revenue
M6	5 (gain-share)	Rs. 600K	—	Rs. 1.8M
M12	20	Rs. 1.7M	—	Rs. 9.6M
M18	75	Rs. 6.4M	Rs. 76M	Rs. 45M
M24	200	Rs. 17M	Rs. 204M	Rs. 160M
M30	450	Rs. 38M	Rs. 459M	Rs. 450M
M36	850	Rs. 72M	Rs. 867M	Rs. 1.1B

Funding Requirements



Seed Round — USD 300K

Use: 3 WattBrain prototypes, 5 pilot installations, 2 engineers + 1 field CSR for 12 months. Enough to prove savings and close first 5 paying customers. Target: LUMS NIC, Invest2Innovate, NLC Innovation.



Series A — USD 2M

Trigger: 50 customers, positive unit economics proven. **Use:** Hardware manufacturing scale, 6-city expansion team, ML team of 3. Target: Sarmayacar, Indus Valley Capital, regional angels.



Series B — USD 10M

Target: 100M ARR, 1000 customers

Trigger: RS. 400M ARR, national coverage.

Use: Bangladesh, Sri Lanka, Egypt expansion.

Energy brokerage license. NEPRA grid data partnership. Target: Tiger Global, Sequoia Southeast Asia.



Path to profitability: WattWise reaches operating breakeven at approximately 120 customers on the Growth plan (~Month 20). At 500 customers, EBITDA margin exceeds 45%. The hardware asset (sensors deployed in factories) appreciates in value as a data moat — each additional month of data makes our ML model more accurate, widening the gap vs. any competitor attempting to enter.

14 Development Roadmap

PHASE 0

Months 1–3

FOUNDATION & PROTOTYPE

Company registration, team hiring, first hardware prototype, cloud infra setup.

- SECP incorporation + NTN + bank account
- Hire: 2 backend engineers, 1 hardware/embedded engineer
- WattBrain v0.1: Raspberry Pi CM4 + 4 CT sensor nodes, Modbus working
- Basic MQTT → InfluxDB pipeline deployed on AWS Bahrain
- Simple dashboard: live kW, historical chart — no ML yet
- Lab testing: simulate load switching, measure accuracy

PHASE 1

Months 4–6

PILOT DEPLOYMENTS (SHADOW MODE)

3–5 factories, sensors only, no automation. Data collection and baseline building.

- Approach APTMA Faisalabad — get 3 pilot factory permissions
- Install WattBrain + 10–20 nodes per factory (shadow mode only)
- Build WAPDA feeder schedule scraper for Faisalabad region
- Train first GOP model on collected data + historical feeder logs
- Run LoadShift optimizer in paper-trade mode (predict, log, don't act)
- Produce Month-1 savings simulation report for each factory

PHASE 2

Months 7–12

LIVE AUTOMATION & FIRST REVENUE

Enable On/Off/Outage control for factories, hosts, meter, shunt, billings

Enable SwiftSwitch on pilot factories, begin gain-share billing.

- SwiftSwitch v1: predictive relay control, 8-second switchover
- Activate automation on 3 pilot factories (with owner sign-off)
- First gain-share invoices issued at Month 8
- Mobile app MVP (Flutter) — factory manager daily summary
- Expand to 15 paying customers by Month 12
- Savings Ledger with SHA-256 audit hash

PHASE 3

Months 13–24

SCALE: FAISALABAD → SIALKOT → LAHORE

Series A funding deployed. Transition from gain-share to subscription. Geographic expansion.

- Annual subscription offering launched (Growth plan)
- Sales team of 5 in Faisalabad + 2 in Sialkot
- WattBrain v2: custom PCB (not Pi), reduced BOM by 40%
- ERP integration APIs (SAP B1, Oracle NetSuite — used by larger mills)
- Carbon emission reporting module (EU buyer compliance)
- Target: 200 customers, Rs. 200M ARR

PHASE 4

Months 25–36

NATIONAL SCALE & PLATFORM PLAY

Energy brokerage, bank partnerships, regional export.

- NEPRA Energy Broker license application (enables demand aggregation)
- Bank API integration: savings data → auto credit scoring for energy upgrades
- WattWise Marketplace: customers trade surplus generator capacity
- Bangladesh pilot (similar load shedding problem, RMG sector = identical ICP)
- Series B raise for regional expansion
- Target: 850 customers, Rs. 867M ARR

15 Team Structure

Founding Team (Required)



CEO / Co-Founder



CTO / Co-Founder

Business development and GTM. Ideally has **textile industry connections** (APTMA contacts, mill owner family background). Sells the first 20 customers personally. Raises funding. Background: MBA or industrial engineering + 3–5 years in Pakistan B2B sales.

Full-stack + IoT. Builds the WattBrain prototype and cloud platform. Must be comfortable in Python (ML), Go (backend), and embedded Linux. Background: CS/EE degree + 2+ years industry.

This role suits your profile, Hammad



Hardware Engineer

PCB design, CT sensor integration, Modbus/RS-485 bus. Handles field installation quality. Ideally has electrical engineering background with hands-on factory experience.

First 12–Month Hiring Plan

Role	Hire Month	Monthly Cost (PKR)	Priority
Backend Engineer (Go + Python)	M1	Rs. 150–200K	CRITICAL
ML Engineer (PyTorch, time-series)	M2	Rs. 150–180K	CRITICAL
Field CSR — Faisalabad (Punjabi speaker)	M3	Rs. 80–100K	HIGH
Frontend Engineer (React)	M5	Rs. 120–150K	MEDIUM
Field CSR — Sialkot	M9	Rs. 80–100K	MEDIUM
Finance / Billing Manager	M10	Rs. 100–120K	MEDIUM



Talent sourcing: NUST, LUMS, PIEAS, GIKI, UET Lahore produce strong CS/EE graduates. Offer equity (1–2% ESOP pool for early engineers) as a retention tool — cash alone won't keep top engineering talent. Consider NEEP/PSEB funded internship programs for junior engineers.

16 Risk Analysis & Mitigation

Risk	Severity	Likelihood	Mitigation
Load shedding improves drastically — reduces market pain	MED	LOW	Historical pattern shows Pakistan's energy crisis is structural, not cyclical. Even partial improvement still leaves Rs. 50–80B in addressable waste. Platform pivots to energy efficiency analytics if grid stabilizes.
Factory owner refuses to share energy data	HIGH	MED	Data stays on-premise on WattBrain. Customer signs data agreement explicitly limiting use. Zero raw data leaves the building — only aggregated savings metrics go to cloud. On-device processing is the selling point.
Counterfeit/clone hardware in market	MED	LOW	Hardware value is low; software and ML model are the moat. Clone hardware without our cloud = no value. Secure boot on WattBrain prevents third-party firmware. Hardware is "free" — eliminates incentive to clone.
WAPDA changes feeder schedule format — scraper breaks	MED	MED	Build real-time voltage/frequency-based prediction as a backup (doesn't rely on schedule). Employ a data ops person to monitor scraper health daily. Explore official WAPDA API partnership after early traction.
Customer disputes savings calculation	HIGH	LOW	Baseline methodology locked in contract before installation. SHA-256 audit hash makes data tamper-evident. Offer an independent audit clause — third party can verify our savings claim. Transparent calculation shown in dashboard.

— USD hardware costs spike in PKR			USD equivalent in contract. Hardware BOM hedged by keeping 3-month inventory buffer. Phase 2 moves to local assembly to reduce import exposure.
Key engineer leaves — IP risk	MED	MED	IP assignment clauses in all employment contracts. Equity vesting over 4 years with 1-year cliff. Code in private GitHub with documented architecture. No single engineer holds all knowledge — cross-training policy.
Security breach — sensor data leaked	HIGH	LOW	Per-device TLS, zero-trust architecture, no raw data in cloud. Penetration test before go-live. Cyber insurance. Incident response plan from Day 1. ISO 27001 audit at Series A.

17 Regulatory & Compliance

Pakistan-Specific Regulatory Landscape



NEPRA Compliance

National Electric Power Regulatory Authority. WattWise is a **behind-the-meter** solution — it does not touch the grid or trade electricity. No NEPRA license required at launch. Monitor evolving demand-side management regulations.



PSQCA Certification

Pakistan Standards & Quality Control Authority certification required for hardware sold/deployed commercially. File for PS:2012 (electrical equipment) certification in Month 3. Timeline: 8–16 weeks.



PECA 2016 (Data Protection)



FBR Tax Structure

Prevention of Electronic Crimes Act. Ensure consent-based data collection, defined retention periods, and right-to-deletion in customer agreements. PTA may introduce formal data localization requirements — AWS Bahrain pre-empts this.

Revenue from SaaS = IT export income if billed in USD to non-resident entities. Otherwise standard corporate tax at 29%. Consult A.F. Ferguson or KPMG Pakistan on IT company classification for 2025 tax year.



Import Regulations

CT sensors: HS 8504.10 (5% customs duty). Microcontrollers: HS 8542.31 (0% under DTRE scheme for exporters). Keep all hardware invoices for customs clearance. Register with WEBOC for streamlined clearance.



Electrical Safety (AEDB)

Alternative Energy Development Board guidelines for behind-the-meter equipment. The SwiftSwitch relay module must be installed by a licensed electrician (C-class license minimum). Keep installation logs per site.



Strategic regulatory play: After Month 18, approach NEPRA with a proposal to be a registered Demand Side Management (DSM) provider. This would let WattWise participate in demand response programs — being paid by the grid to reduce industrial load during peak demand. This is a business model layer that could be worth as much as the SaaS itself.

Key Contracts Required

-  **Customer Agreement:** Defines baseline methodology, savings calculation rules, gain-share percentage, dispute resolution, data ownership, hardware return policy. Reviewed by a Pakistani commercial lawyer. Non-negotiable standard terms for Starter plan.
-  **Data Processing Agreement:** PECA-compliant. Specifies what data is collected, where stored, retention period (3 years for savings audit), and customer's right to export or delete.
-  **Hardware Deployment Agreement:** Hardware remains WattWise property. Customer provides access, power, and network. Hardware returned if subscription lapses.
-  **Employee IP Assignment:** All IP created during employment assigned to WattWise. Non-compete for 1 year in the energy intelligence space. Non-solicit for 2 years.

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